Closure of Attributes in Relational Databases

1 Closure of Attributes

1.1 What is Attribute Closure?

Definition

The **closure** of a set of attributes X, written as X^+, is the complete set of attributes that can be determined from X using the given functional dependencies.
In other words, if you know the values of attributes in X, then X^+ tells you all other attributes you can find out based on the rules.

1.2 Why is it Important?

- To check if X is a **superkey** — meaning X can uniquely identify every record in the table.
- To find out if a certain rule (functional dependency) $X \to Y$ holds based on existing rules.
- To help organize and improve the design of the database, reducing redundancy.

1.3 How to Calculate X^+

1. Start with all attributes in X:
$$X^+ = X$$
2. Look at each functional dependency $Y \to Z$:
   - If all attributes in Y are already in X^+, then add all attributes in Z to X^+.
3. Repeat step 2 until no new attributes can be added.

1.4 Example

Example: Finding Closure of AG

Suppose a table has attributes:

$$R = (A, B, C, G, H, I)$$

and functional dependencies:

$$F = \{A \to B, \quad A \to C, \quad CG \to H, \quad CG \to I, \quad B \to H\}$$

Calculate the closure of AG, i.e., $(AG)^+$:

- Start with: $(AG)^+ = \{A, G\}$
- $A \to B$ applies because $A \in (AG)^+$, add B:

$$(AG)^+ = \{A, G, B\}$$

- $A \to C$ applies, add C:

$$(AG)^+ = \{A, G, B, C\}$$

- $B \to H$ applies, add H:

$$(AG)^+ = \{A, G, B, C, H\}$$

- Since both C and G are in $(AG)^+$, $CG \to I$ applies, add I:

$$(AG)^+ = \{A, G, B, C, H, I\}$$

Now, $(AG)^+$ contains all attributes in R. Therefore, AG can determine every attribute in the table.

1.5 What Does This Tell Us?

- AG is a **superkey** because it can uniquely identify every record.
- Check smaller subsets:
  - $A^+ = \{A, B, C, H\}$ (missing G, I) — not a superkey.
  - $G^+ = \{G\}$ only — not a superkey.
- So, AG is also a **candidate key** (a minimal superkey).

1.6 Why Do We Use Attribute Closure?

Uses of Attribute Closure

Attribute closure is very helpful when working with databases. Here are three main reasons why we use it:

- **Checking for Superkeys:** A *superkey* is a set of columns (attributes) that can identify each row in a table uniquely. To check if a set X is a superkey, we find its closure X^+. If X^+ has all the columns in the table, it means X can find out everything, so it is a superkey.
  Example: If $(AG)^+$ contains all columns, then AG is a superkey.
- **Checking Functional Dependencies:** A functional dependency $X \to Y$ means knowing X lets us find Y for sure. To check if this is true, we find X^+. If all columns in Y are in X^+, then the rule $X \to Y$ is correct.
  Why? This helps us confirm the rules or relationships in the data before building the database.
- **Making Database Design Better:** We want to avoid storing the same data more than once, which can cause mistakes. Using attribute closure helps us:
  - Find the keys (like the main ID) that identify rows.
  - Find extra rules that cause repeated data.
  - Organize tables to remove repeated or unnecessary data.

  This keeps the database clean, simple, and easy to update.

Solved Examples

Example 1: Computing Attribute Closure

Example: Compute $(A)^+$ for $R = (A, B, C, D, E)$

Given relation schema:

$$R = (A, B, C, D, E)$$

and functional dependencies:

$$F = \{A \to BC, \quad CD \to E, \quad B \to D, \quad E \to A\}$$

Compute the closure of attribute set A.

Solution:

- Start with $A^+ = \{A\}$.
- Apply $A \to BC$:
$$A^+ = \{A, B, C\}$$
- Apply $B \to D$ (since $B \in A^+$):
$$A^+ = \{A, B, C, D\}$$
- Apply $CD \to E$ (since $C, D \in A^+$):
$$A^+ = \{A, B, C, D, E\}$$
- $E \to A$ is already satisfied (since $A \in A^+$).

Since $A^+ = \{A, B, C, D, E\} = R$, the attribute set A is a **superkey**.

Example 2: Candidate Keys Identification

Example: Find Candidate Keys for $R = (A, B, C, D, E)$

Given:

$$F = \{A \to BC, \quad CD \to E, \quad B \to D, \quad E \to A\}$$

Find all candidate keys for R.

Solution:

- $A^+ = R$ (from previous example), so A is a candidate key.
- E^+:

  $$E \to A \implies E^+ = \{E, A\} \implies A \to BC \Rightarrow E^+ = \{A, B, C, E\}$$

  $$B \to D \implies E^+ = \{A, B, C, D, E\} = R$$

  So E is also a candidate key.
- CD^+:

  $$CD \to E \implies CD^+ = \{C, D, E\} \implies E \to A \Rightarrow CD^+ = \{A, C, D, E\}$$

  $$A \to BC \implies CD^+ = R$$

  So CD is a candidate key.
- BC^+:

  $$B \to D \implies BC^+ = \{B, C, D\}$$

  $$CD \to E \implies BC^+ = \{B, C, D, E\}$$

  $$E \to A \implies BC^+ = R$$

  So BC is also a candidate key.

Candidate keys are:

$$\boxed{A, \quad E, \quad CD, \quad BC}$$